

HDOC - C Programming Activity

Name	Gurveer Singh Cheema
SAF ID	590032925
Batch	15

Question

Write a C program to perform the following menu-driven operations on a given array:

1. Sort the elements using Bubble Sort algorithm and perform a Binary Search.
2. Perform Insertion and Deletion at a given valid position.
3. Rotate the array elements to the right by k positions.
4. Exit.

Program

```
#include <stdio.h>

#define MAX 100

void display(int a[], int n) {
    int i;
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

void bubbleSort(int a[], int n) {
    int i, j, temp;
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
}

int binarySearch(int a[], int n, int key) {
    int low = 0, high = n - 1, mid;
    while (low <= high) {
        mid = (low + high) / 2;
        if (a[mid] == key)
            return mid;
        else if (a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

int main() {
    int a[MAX], n, choice, i, key, result;
    int pos, value, k, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    do {
        printf("\n--- MENU ---\n");
        printf("1. Bubble Sort and Binary Search\n");
```

```

printf("2. Insertion and Deletion\n");
printf("3. Right Rotation by k positions\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
case 1:
    bubbleSort(a, n);
    printf("Sorted array: ");
    display(a, n);

    printf("Enter element to search: ");
    scanf("%d", &key);
    result = binarySearch(a, n, key);

    if (result != -1)
        printf("Element found at position %d\n", result + 1);
    else
        printf("Element not found\n");
    break;

case 2:
    printf("Enter position and value for insertion: ");
    scanf("%d %d", &pos, &value);

    if (pos < 1 || pos > n + 1 || n >= MAX) {
        printf("Invalid insertion position\n");
        break;
    }

    for (i = n; i >= pos; i--)
        a[i] = a[i - 1];
    a[pos - 1] = value;
    n++;

    printf("After insertion: ");
    display(a, n);

    printf("Enter position for deletion: ");
    scanf("%d", &pos);

    if (pos < 1 || pos > n) {
        printf("Invalid deletion position\n");
        break;
    }

    for (i = pos - 1; i < n - 1; i++)
        a[i] = a[i + 1];
    n--;

    printf("After deletion: ");
    display(a, n);
    break;

case 3:
    printf("Enter k: ");
    scanf("%d", &k);

    if (n > 0) {
        k = k % n;
        while (k--) {
            temp = a[n - 1];
            for (i = n - 1; i > 0; i--)
                a[i] = a[i - 1];
            a[0] = temp;
        }
    }

    printf("Array after right rotation: ");
    display(a, n);
    break;

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice\n");
}

```

```
    }  
    } while (choice != 4);  
  
    return 0;  
}
```

Sample Output

```
Enter number of elements: 5  
Enter 5 elements: 40 10 30 20 50
```

```
--- MENU ---  
1. Bubble Sort and Binary Search  
2. Insertion and Deletion  
3. Right Rotation by k positions  
4. Exit  
Enter your choice: 1  
Sorted array: 10 20 30 40 50  
Enter element to search: 30  
Element found at position 3
```

```
Enter your choice: 2  
Enter position and value for insertion: 3 25  
After insertion: 10 20 25 30 40 50  
Enter position for deletion: 4  
After deletion: 10 20 25 40 50
```

```
Enter your choice: 3  
Enter k: 2  
Array after right rotation: 40 50 10 20 25
```

```
Enter your choice: 4  
Exiting program...
```

Result: The menu-driven C program successfully performs Bubble Sort with Binary Search, insertion, deletion, and right rotation of the array.